RESEARCH BRIEFING

Aligning OBIS's new indexing model with GBIF's data representation

Verbatim vs interpreted fields, and how extension records are named and nested

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Context: OBIS `separate-tables` Parquet prototype, Elasticsearch redesign, GBIF Pipelines / occurrence API / download formats, DwC-A extensions registry

Prepared 12 October 2026

1 Executive summary

OBIS is redesigning how DwC-A data are stored and served: separate Parquet tables for events, occurrences, and extensions (the `separate-tables` branch), plus a leaner Elasticsearch surface that no longer densifies event-level MoF onto every occurrence. That redesign is not only a scaling fix. It is a rare chance to **align the OBIS internal model with GBIF's** on two points that matter for shared tooling, dual-network publishing, and user expectations:

1. **Verbatim vs interpreted** — keep publisher strings separate from typed, QC'd, taxonomy-matched values.
2. **Extension containers** — stop using OBIS-only short names (`mof`, `dna`) as the **canonical** storage/API shape; prefer GBIF's `extensions` map keyed by extension **rowType** IRI (with optional short aliases for the OBIS API and R package).

The prototype already gestures at (1) with top-level `source` and `interpreted` structs. The recommendation below is to push that further toward GBIF naming and semantics, while keeping OBIS-specific QC (`dropped`, `absence`, bathymetry enrichments) as first-class companion fields — not mixed into verbatim.

> **Bottom line**
>
> Use the new Parquet + ES indexes to adopt GBIF's **split**: `verbatim` (IRI-keyed or local-name maps of publisher strings) + `interpreted` (typed OBIS/WoRMS view) + `extensions` (rowType-keyed, **directly linked only** on occurrence documents). Keep multi-table Parquet as the download truth for event-level extensions. Do not wait for DwC-DP to make this change — it helps today's DwC-A stack and leaves the door open for DwC-DP landers later.

2 Why alignment matters now

2.1 What is changing in OBIS

Today's production path is occurrence-centric: Event-core archives are flattened so occurrences inherit empty core fields **and** parent event extension rows. Elasticsearch documents expose a flat bag of Darwin Core–ish fields plus nested `mof` / `dna` arrays (often including `level` markers for event inheritance). That made the R package and simple API clients easy — and created mass duplication when stations carry dozens of sediment parameters.

The new model (documented in the project README and `diagrams/indexing.svg`) keeps:

- **Field inheritance** onto interpreted occurrence/event rows (so filters on latitude or date still work).
- **Extension rows in separate tables** (occurrence vs event MoF / DNA), joined at download time via `(dataset_id, …ID)`.

Elasticsearch should then nest only **directly linked** extensions on occurrences (DNA is the obvious case), matching how GBIF already behaves for the same Antarctic fish record: one length MoF on the occurrence in GBIF; many event-level MoF rows nested under the occurrence in OBIS today.

2.2 What GBIF already does

GBIF's public processing story is three layers (GBIF infrastructure: data processing):

Layer	Role
Fragment	Raw crawl payload (DwC-A row, ABCD, …) as received.
Verbatim (`ExtendedRecord`)	Normalised DwC star: core terms + map of extensions; values still **strings**; terms often stored as IRIs.
Interpreted	Typed fields, controlled vocabularies, taxonomy match, coordinates/issues; category Avro files in Pipelines; ES / API / `occurrence.txt` downloads.

Table 1: GBIF occurrence processing layers (simplified).

The occurrence API returns the **interpreted** view with an `extensions` object keyed by extension rowType IRI. A separate `/verbatim` view (and download `verbatim.txt` + optional verbatim extension files) preserves publisher strings (GBIF download formats). Core identity uses `gbifID` structurally while `occurrenceID` remains publisher data — the same structural/semantic split later formalised for DwC-DP (#159).

OBIS does not need to copy Spark, Avro category files, or `gbifID` minting to benefit from this **shape**.

3 Gap analysis: OBIS today vs GBIF

3.1 Verbatim vs interpreted

Aspect	GBIF	OBIS (production / prototype)
Publisher strings	`/verbatim` API; `verbatim.txt`; IRI-keyed core terms in `ExtendedRecord`	Mostly lost at API surface; prototype Parquet has `source` struct
Processed values	Typed API fields; `occurrence.txt`; issues list	Flat API fields (`decimalLatitude`, `aphiaID`, `flags`, …); prototype `interpreted` struct
Taxonomy	GBIF Backbone (+ multi-classification payloads)	WoRMS / Aphia IDs and ranks on the document
QC / enrichment	`issues`, media, facts, …	`dropped`, `absence`, `bathymetry`, `sst`, `sss`, `shoredistance`, node tags
Naming of raw layer	`verbatim`	`source` (prototype) — easy to rename

Table 2: How publisher vs processed values appear today.

The prototype schema already separates concerns:

```
{
  "dataset_id": "…",
  "source": { "…publisher fields…" },
  "interpreted": { "eventID": "…", "occurrenceID": "…", "…" }
}
```

That is closer to GBIF than the flat production ES document. The remaining gaps are mostly **naming**, **term keying**, and **what is allowed inside each struct** (do not put OBIS enrichments into `source` / `verbatim`).

3.2 Extension record containers and field names

Aspect	GBIF	OBIS today
Top-level key	`extensions`	`mof`, `dna` (and similar short names)
Extension identity	RowType IRI, e.g. `http://rs.iobis.org/obis/terms/ExtendedMeasurementOrFact`	Implied by property name
Term keys inside a row	Usually full IRIs in verbatim/API extensions	Local names (`measurementType`, …)
Event MoF on occurrence	Not densified	Densified with `level` (0 = occurrence, 1 = event, …)
Downloads	Separate extension files / `verbatim/*.txt` when requested	Historically nested in occurrence export; prototype: separate Parquet tables

Table 3: How extension rows are exposed.

Concrete contrast for the same occurrence (`gbifID` / shared dataset content around Notolepis coatsi, July 2006):

- **GBIF API** — `extensions["http://rs.iobis.org/obis/terms/ExtendedMeasurementOrFact"]` with one length measurement; IRI-keyed fields.
- **OBIS API** — `mof: [ … many rows … ]` with local names and `level`, including station/net parameters inherited from the event.

The separate-tables prototype already **stores** event and occurrence MoF apart (`event_measurementorfact.parquet` vs `occurrence_measurementorfact.parquet`). Aligning the **API and ES field names** with GBIF is the remaining product choice.

3.3 Field inheritance (related, not identical)

OBIS still fills empty occurrence core fields from parent events in the interpreted view (needed for geographic/temporal filters). GBIF's behaviour is more conservative for some Event-core datasets (coordinates/depth may remain on the event only — see pipelines #878 and the Humboldt hierarchy notes). Recommendation for OBIS: keep **interpreted** field inheritance for search; do **not** inherit extension rows; document that `verbatim` / `source` reflects the publisher row **before** inheritance.

4 Target model for the new indexes

4.1 Parquet lake (Track B — downloads and analytics)

Adopt a GBIF-inspired row shape for every core and extension table:

```
{
  "dataset_id": "5adfc524-…",
  "verbatim": { },
  "interpreted": { },
  "issues": [ ],
  "obis": { "dropped": false, "absence": false, "node_ids": [], "…" }
}
```

> **Naming**
>
> Prefer renaming prototype `source` → `verbatim` in new exports. Keep a temporary dual-read (`source` alias) for one migration cycle if needed. Do not invent a third name.

Conventions:

- **`verbatim`** — publisher strings only; prefer **stable local Darwin Core / extension term names** in Parquet for DuckDB ergonomics, and retain a mapping table (local name ↔ IRI) shared with GBIF's TermFactory list where possible. Optionally store IRI keys in a parallel `verbatim_iri` map if round-tripping to GBIF-shaped JSON matters.
- **`interpreted`** — typed or normalised values OBIS is willing to filter and display; includes inherited event fields when OBIS chooses to inherit; includes WoRMS matches.
- **`issues` / flags** — machine-readable QC, analogous to GBIF `issues` (OBIS can keep existing flag vocabularies under a clear namespace).
- **`obis` (or top-level OBIS columns)** — enrichments that are not Darwin Core (`bathymetry`, `sst`, …). Keeping them out of `interpreted` DwC keys avoids pretending they were published.

Extension tables stay separate files (as now), each row with the same verbatim/interpreted envelope and composite join keys `(dataset_id, occurrenceID|eventID)`. Table file names can stay snake_case for the lake (`occurrence_measurementorfact.parquet`) while **API** uses rowType IRIs.

4.2 Elasticsearch / API (Track A — search)

Mirror GBIF's public occurrence document **shape** without giving up OBIS fields:

Block	Content
Identity	`id` / `obisID`, `dataset_id`, `occurrenceID`, `eventID`, …
Interpreted DwC	Flattened searchable fields (as today), clearly documented as interpreted
OBIS enrichments	`bathymetry`, `sst`, `dropped`, …
`extensions`	Map of rowType IRI → array of term maps; **only directly linked** rows
`extensionsShort` (optional)	Aliases `mof` / `dna` for backward-compatible API clients — derived views, not a second store
`verbatim` (optional on ES)	Omit or store sparsely; full verbatim lives in Parquet / a `/verbatim` style endpoint if needed

Table 4: Proposed occurrence search document (conceptual).

Event index documents follow the same pattern: event-level MoF/DNA live under the **event** document's `extensions`, not copied onto child occurrences.

Priority extensions (from the project README) should register under their GBIF/TDWG rowType IRIs from day one of the new index:

- Measurement or Facts / eMoF — `http://rs.tdwg.org/dwc/terms/MeasurementOrFact` or `http://rs.iobis.org/obis/terms/ExtendedMeasurementOrFact` (pick one canonical per pipeline version; map the other on read if both exist in the wild).
- DNA derived data — `http://rs.gbif.org/terms/1.0/DNADerivedData`
- Simple Multimedia — `http://rs.gbif.org/terms/1.0/Multimedia`
- Audiovisual / Identification History / Humboldt — as registered on rs.gbif.org

4.3 Download packaging

Align **user-facing** download layouts with GBIF vocabulary even when the bytes are Parquet:

GBIF download	OBIS Track B analogue
`occurrence.txt` (interpreted)	`occurrence.parquet` → `interpreted` (+ flattened CSV option)
`verbatim.txt`	`occurrence.parquet` → `verbatim` (CSV/JSONL projection)
Extension files / `verbatim/*.txt`	`occurrence_*` / `event_*` extension Parquets; optional IRI-keyed JSON
DwC-A zip with `id`/`coreid`	Future packager: remint structural keys; see companion briefing on #159 / composite lake keys

Table 5: Conceptual mapping of GBIF download files → OBIS multi-table export.

Document explicitly: event-level measurements are **not** repeated under each occurrence in either Parquet or default API payloads.

5 Implementation sketch (stay close to GBIF, stay runnable on DuckDB)

5.1 Pipeline stages

1. **Land** DwC-A (and later DwC-DP) into per-dataset table folders — already the `separate-tables` direction.
2. **Build verbatim maps** from core + extension files (stringly; preserve empty vs absent if possible).
3. **Interpret** into typed fields + WoRMS + OBIS enrichments; apply **field** inheritance here only.
4. **Write Parquet** with `verbatim` / `interpreted` / OBIS blocks; extension tables without cross-occurrence copy.

5. **Project ES documents** from interpreted + **direct** extensions only; attach `extensions` as rowType map.
6. **Downloads** = DuckDB key walk over Parquet (existing `query_subset.py` pattern).

5.2 Term keys: IRI vs local name

GBIF verbatim JSON leans on IRIs; DuckDB users prefer `measurementType`. Pragmatic compromise:

- **Parquet lake:** local names inside `verbatim` / `interpreted` structs (current prototype style).
- **API `extensions` objects:** IRI keys for extension **types**; local names **or** IRIs for fields inside rows — if only one, prefer local names plus a published term list, or emit both (`measurementType` and the IRI) during a transition.
- **Interop export** aimed at GBIF-shaped consumers: a small serializer that rewrites local names → IRIs using a shared dictionary (the same list GBIF's TermFactory knows).

5.3 Backward compatibility

Breaking `mof` / `dna` will hurt the R package and existing scripts. Mitigations:

1. API flag `extensions=true` returns GBIF-shaped map; default retains `mof`/`dna` aliases for one major version.
2. R package gains download helpers for multi-table subsets (event MoF via join) while keeping "quick DNA on occurrence" via nested direct extensions.
3. Dual-write `source=verbatim` in Parquet until downstream jobs migrate.

6 What not to copy from GBIF

- Spark / HDFS / category Avro file sprawl — DuckDB + Parquet is enough at OBIS scale.
- Blind adoption of GBIF Backbone as the only taxonomy — keep WoRMS; optionally **add** a GBIF backbone classification block later for crosswalk, as GBIF's multi-classification payload shows is possible.
- Densifying DwC-DP graphs into occurrence documents — the companion cross-dataset briefing already argues for bounded projection.
- Using Darwin Core ID fields as the only join keys inside multi-dataset Frictionless zips — composite `(dataset_id, …)` in the lake; remint `_pk`/`_fk` at package time.

7 Recommendations

1. **Treat the indexing redesign as an alignment project**, not only a densify-removal project. Freeze a target document schema before rebuilding production ES.
2. **Rename `source` → `verbatim`** in new Parquet exports; keep `interpreted`; park OBIS-only enrichments outside DwC maps.
3. **Canonical extension container = `extensions` keyed by rowType IRI** in ES/API; provide `mof`/`dna` aliases temporarily.
4. **Nest only directly linked extensions** on occurrence search documents; serve event extensions from the event index or from Parquet downloads.
5. **Preserve verbatim for provenance and debugging** (Parquet always; API `/verbatim` or include-on-demand later).
6. **Publish a short term/extension dictionary** (local name ↔ IRI ↔ table file) so OBIS and GBIF tooling can share serializers.

7. **Sequence work as:** Parquet shape → download/API contracts → ES mapping → R package — so clients move once.

> **Decision needed**
>
> Confirm the canonical eMoF rowType string for OBIS indexes (`rs.iobis.org` ExtendedMeasurementOrFact vs TDWG MeasurementOrFact) and whether API extension **field** keys go IRI-first or local-name-first. Everything else in this briefing can proceed either way if the choice is explicit and versioned.

8 Conclusions

The new OBIS indexes already move toward GBIF's architecture by splitting tables and stopping extension densification. Closing the remaining gap — **verbatim vs interpreted envelopes** and **GBIF-like `extensions` naming** — is mostly convention and API design, not a new storage platform. Doing it now reduces future DwC-DP friction, improves dual-network mental models for users, and makes shared validation/export utilities realistic. Composite lake keys and optional structural keys for packages remain as in the companion DwC-DP cross-dataset briefing; they complement, rather than replace, this representation alignment.

9 References

1. OBIS DwC-DP / indexing notes — project `README.md` (draft), `diagrams/indexing.svg`.
2. iobis/pipeline `separate-tables` — multi-table Parquet export prototype.
3. Local subset exporter — `scripts/query_subset.py`.
4. GBIF infrastructure: data processing — fragment / verbatim / interpreted.
5. GBIF occurrence download formats — `occurrence.txt`, `verbatim.txt`, extension files.
6. gbif/pipelines — `ExtendedRecord`, interpretation to category Avro, ES indexing.
7. GBIF occurrence API example and `/verbatim` sibling.
8. OBIS occurrence API — flat fields + `mof` / `dna`.
9. gbif/pipelines#878 — event hierarchy / inheritance discussion.
10. Humboldt hierarchical events — event levels and inheritance expectations.
11. GBIF extensions registry — MoF, DNA, Multimedia, Humboldt, …
12. Companion briefing — `reports/gbif-cross-dataset-dwcdp.typ` (structural vs semantic keys; OBIS Track A/B).

This briefing synthesizes OBIS prototype work and public GBIF processing/API documentation as of 12 October 2026. It proposes representation alignment for the new Parquet and Elasticsearch indexes; it does not commit OBIS to GBIF's taxonomy backbone or compute stack. Typographic report generated with Typst.